

ARMIN RAHIMI

AI Engineer · Speech & LLM & Infrastructure

PROFESSIONAL SUMMARY

AI Engineer specializing in production-grade Speech AI (ASR / TTS / WUW) and Large Language Models. Trained models on 80,000+ hours of audio data, deployed multi-GPU inference infrastructure on H100s, and reduced system latency by 40% via TensorRT optimization. Expert in end-to-end ML pipelines, from data engineering and model training to architecting real-time streaming APIs and multi-agent LLM systems. Hands-on with full-stack infrastructure: provisioned and managed KVM-based VMs from scratch, containerized services with Docker, configured Nginx reverse proxies, tracked model experiments and deployments with MLflow, and centralized observability via Grafana dashboards and Kibana log aggregation. Comfortable with CI/CD workflows and Linux server administration across all stages of the AI product lifecycle.

WORK HISTORY

AI Engineer · Jun 2024 – May 2026

Smilinnno · Tehran, Iran

Speech Recognition (ASR)

- Fine-tuned ASR models on 40,000+ hours of Persian speech utilizing PyTorch and Hugging Face.
- Built a production-grade streaming ASR pipeline achieving sub-second latency.
- Developed a Telephony Simulation Pipeline that converted 30,000+ hours of data into narrowband audio.
- Transcribed 10,000+ hours of real phone call data to significantly enhance model robustness.
- Engineered a specialized, noise-robust ASR model optimized for narrowband telephony, mitigating signal distortion in challenging environments.

Text-to-Speech (TTS)

- Deployed a bilingual (FA/EN) TTS engine trained on 40,000 hours of audio data.
- Reduced inference latency through custom TensorRT kernel optimization.
- Implemented automated Persian Text Normalization pipelines for numbers, dates, and complex formulas.

Wake Word Detection (WUW)

- Designed custom low-latency Neural Network for Wake Word Detection.
- Trained on 5M+ balanced samples from 4,300+ natural speakers
- Engineered a few-shot learning engine to support user-defined trigger phrases.
- Deployed TFLite/ONNX model onto Android (Zigap app), achieving high accuracy with minimal power consumption.

LLM & Agentic Systems

- Designed advanced RAG pipelines for enterprise document Q&A using Chroma vector databases.
- Fine-tuned LLaMA-family models with LoRA/QLoRA techniques for domain-specific tasks.
- Architected Multi-Agent "Judge" systems to improve SQL generation accuracy and reliability.
- Deployed high-performance LLM inference environments using vLLM and NVIDIA Triton on dual H100 GPUs.

Inference Infrastructure

- Achieved 40% system-wide latency reduction via TensorRT model compilation.
- Built asynchronous Worker-Queue Architecture (RabbitMQ / Redis / Celery) for bulk audio processing.
- Real-time WebSocket streaming pipelines with FastAPI.



CONTACT

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SKILLS

- PyTorch · TensorFlow · HuggingFace
- ASR · TTS · Wake Word (WUW)
- LLM Fine-tuning (LoRA / QLoRA)
- RAG · Vector DBs · NLP
- NVIDIA Triton · TensorRT · vLLM
- Multi-GPU (H100) · CUDA
- Docker · KVM · Linux · Nginx
- FastAPI · WebSockets · Redis
- MLflow · Grafana · Kibana · CI/CD
- TFLite · ONNX · Edge AI
- Python · C++ · Bash

EDUCATION

M.Sc. Medical Information Technology Engineering
University of Tehran
2021 – 2023

B.Sc. Electrical Engineering
University of Kurdistan
2013 – 2018

CERTIFICATIONS

- Machine Learning Specialization
Andrew Ng · Coursera (3 courses)

LANGUAGES

- Persian — Native
- English — Intermediate
- Kurdish — Native

- Provisioned and managed KVM virtualization with GPU passthrough across isolated production VMs dedicated to ASR, TTS, and model training workloads — built and configured the full stack from scratch.
- Configured Nginx as a reverse proxy for routing traffic across multiple AI services.
- Tracked model training experiments, logged metrics, and managed model registry using MLflow.
- Built Grafana dashboards for real-time monitoring of inference services and system health.
- Centralized service logs in Kibana (ELK stack) for structured log aggregation and debugging.
- Applied foundational CI/CD practices (Git-based workflows, automated deployment pipelines) for iterative service delivery.

PROJECTS

Knee Osteoarthritis Detection System

Developed an end-to-end deep learning pipeline for automated X-Ray analysis, utilizing CLAHE preprocessing, feature extraction, and classification.

Enterprise Social Network Platform

Implemented real-time organizational feeds with WebSocket updates and a custom gamification engine featuring virtual currencies, peer-to-peer recognition, and achievement tracking.